Hierarchical Hybrid Planning for Multi-AGV Scheduling in Multi-Floor Factories

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Abstract—Multi-floor manufacturing requires automated guided vehicles (AGVs) to share both floor routes and lift resources. We reimplement the prioritized multi-AGV scheduling baseline of Guo et al. and evaluate a hierarchical extension in which a central planner assigns jobs and books lifts while floor-level planners schedule local route legs using lift lobbies as waiting points. In the tested three-floor layout, the hybrid configuration reduces mean makespan by 65%, total tardiness by 72%, and load imbalance by 80% relative to the paper-style centralized configuration over ten job sets of 100 jobs and 20 AGVs. A centralized FCFS-leg ablation produces the same schedules as the hybrid on the tested seeds, so the measured schedule gains are associated with lobby waiting and FCFS leg ordering; floor decomposition changes where reservation state and computation are held. Five-floor sensitivity and finite-lobby experiments show that outcomes depend on layout and capacity-handling choices, with some centralized finite-capacity runs failing verification. All reported successful plans pass an independent checker. These simulation results apply to the tested layouts, seeds, and parameters; they do not establish a general scalability or optimality claim.

Index Terms—Automated guided vehicles, multi-agent path finding, prioritized planning, multi-floor logistics, lift coordination, hierarchical scheduling.

I. Introduction

Automated guided vehicles (AGVs) are a central part of material handling in smart manufacturing. They move workpieces between storage, processing, and delivery locations, allowing production systems to respond flexibly to changing orders and workloads. As factories grow, however, coordinating a fleet becomes a joint assignment, routing, and traffic-management problem. Vehicles may need the same narrow aisle, workstation approach, or transport resource at overlapping times. A schedule must therefore assign jobs and produce routes and timings that avoid collisions while meeting operational targets such as completion time, delivery due dates, and balanced vehicle utilization. These requirements become more demanding in multi-floor facilities, where lifts connect otherwise separate floor networks. A lift is a shared resource with finite capacity and non-negligible travel time: a conflict at one lift can delay routes on several floors and propagate through the entire schedule.

Many AGV scheduling methods centralize assignment and traffic planning. This provides a system-wide view, but it also concentrates computation and reservation state at one server. As the fleet and route set grow, that server must reason about an increasing number of node and track occupancy windows, including interactions among vehicles operating in different areas. The planning workload and shared state can consequently become a bottleneck. This challenge is especially relevant to multi-floor factories: a central planner must coordinate local floor traffic as well as cross-floor lift movements. A practical architecture should retain enough central coordination to prevent conflicting lift use while distributing the detailed routing and traffic state across the factory.

Guo et al. present a prioritized planning method for large-scale multiple-AGV scheduling in smart manufacturing [1]. Their approach assigns transport tasks and plans AGV routes sequentially according to priority, resolving conflicts with respect to plans already made for higher-priority vehicles. This provides the centralized baseline for the present work. In that baseline, assignment and conflict-free routing are handled by a single planning process [1].

This paper extends that prioritized-planning baseline with a hybrid central-plus-edge architecture for a multi-floor factory. A central server performs job assignment and divides each AGV route into floor-level legs. It also books the lift trips, which remain shared across floors and therefore require system-wide coordination. Each floor has an edge server that plans its local route legs and resolves node and track conflicts using the prioritized conflict-free routing procedure. The decomposition limits each floor server's detailed reservation state to its own floor, while the central server retains responsibility for assignments and inter-floor transitions. To give vehicles a safe place to wait near a local conflict or a busy lift, the layout adds a lobby node at each lift landing. A floor server can delay a leg at its entry lobby, and the central server can schedule a lift departure after the lift resource becomes available. These lobbies provide waiting points near the conflict rather than requiring every delay to be absorbed at the factory's single parking node.

The implementation is evaluated against two centralized configurations: the paper-style planner, and a centralized planner with the same lift lobbies as the hybrid system. The saved comparison uses 100 jobs and 20 AGVs over 10 random seeds; each generated plan is checked by an independent verifier. In the reported mean results, the hybrid configuration reduces last-delivery makespan by

65%, total tardiness by 72%, and usage-time imbalance by 80% relative to the paper-style centralized configuration. Its busiest-server planning CPU time is also lower. These figures describe the project's modeled factory, parameters, and tested seeds; they should be read as experimental results for this setup, not as general guarantees for all multi-floor factories. The comparison also includes a fleet-size sweep over 4, 8, 12, 16, and 20 AGVs, providing a view of how the measured scheduling metrics vary with fleet size [2].

The remainder of the paper is organized as follows. Section II reviews related work; Section III defines the scheduling model; Section IV describes the prioritized and hybrid methods; Section V gives the experimental setup; Sections VI and VII report performance and fairness results; Section VIII discusses the scope and limitations; and Section IX concludes the paper.

Contributions

- Reimplementation of the ICARCV 2024 prioritized-planning baseline, including fixes to the original demo's documented bugs.
- A hybrid architecture that separates central job assignment and lift booking from per-floor conflict-free route planning.
- Lift-lobby waiting points that allow delays to be absorbed near floor-level conflicts and lift resources.
- Independent verification of completed plans for node and track conflicts, edge travel times, dwell windows, route endpoints, and exactly-once job assignment.
- Comparative results over ten random seeds and a fleet-size sweep.

II. Related Work

A. Centralized prioritized planning and conflict search

Guo et al.'s ICARCV 2024 method is the baseline for this study. It formulates large-scale multiple-AGV scheduling as prioritized assignment and route planning, resolving conflicts against previously planned higher-priority vehicles [1]. The sequential approach avoids joint search over all vehicle paths, but its result depends on assignment and priority order. Prioritized wait adjustment instead starts from individual paths and shifts waiting using safe intervals; its preliminary evaluation reports 50–250 agents and trades solution quality for runtime [3].

Conflict-based search (CBS) offers a coupled alternative. Continuous-time CBS methods model nonuniform action durations and geometric collisions [4]. He et al. extend CBS to multi-floor MAPF with single-agent elevator occupancy and elevator reset intervals, explicitly including lift conflicts in search [5]. These methods provide stronger conflict reasoning, while our experiments focus on job assignment, prioritized route windows, and lift arbitration in a decomposed planner.

B. Distributed multi-robot and AGV coordination

Distributed manufacturing schedulers assign decisions to resource agents. Zhou et al. use collaborative reinforcement-learning schedulers for heterogeneous smart-factory units and consider rush orders and machine failures [6]. Maoudj et al. formulate a capacitated multi-AGV scheduling problem with conflicting products and compare a decentralized multi-agent approach with a mixed-integer model [7]. Draganjac et al. present decentralized traffic management based on local coordination for free-ranging industrial vehicles [8]. These works address decentralized decisions or online manufacturing dynamics; they do not evaluate the same combination of a central lift calendar and floor-local route-leg reservations used here.

C. Hierarchical and zone-based planning

Spatial decomposition reduces the planning scope by assigning agents to regions or layers. The spare-zone method of Zhao et al. uses a hierarchical allocation layer and local coordination to resolve collisions and deadlocks for vehicles following fixed paths [9]. Pratissoli et al. use a centralized, decoupled, multilayer traffic manager with topological sectors, traffic-aware route planning, and time-expanded coordination in an industrial setting [10]. HMAPP combines high-level region sequences with regional conflict-based searches [11]; hierarchical large-scale multirobot replanning similarly combines global congestion-aware routing and parallel low-level planning [12]. Verma et al.'s dynamic resource reservation method allows multiple reservations at shared resource points and can be implemented centrally or with local controllers [13]. These approaches motivate layered coordination and local waiting or conflict-resolution areas. The present implementation instead partitions local route windows by floor while retaining central job assignment and lift booking.

D. Learning-based and optimization-based AGV scheduling

Scheduling quality and traffic efficiency have also been addressed with optimization, learning, and large-neighborhood methods. Wang et al. use a neural network to predict travel time in a multi-state AGV scheduling method for flexible manufacturing [14]. Lu et al. combine workstation assignment rules with a space-time scheduling model and cooperative optimization for conflict-free AGV routes [15]. Hu et al. combine reservation timetables, task allocation, and variable-neighborhood search for multi-load AGVs; their experiments include fleets from 3 to 200 vehicles [16]. MAPF-LNS2 repairs collisions by repeatedly replanning subsets of agents and reports results up to 8,000 agents on a congested warehouse benchmark under a one-hour limit [17]. These approaches target search quality, assignment, or scale, but the cited formulations do not evaluate our specific central-lift and floor-server split.

E. Multi-floor warehouses and lifts

Elevator-aware warehouse routing is studied directly by Ma et al. for a double-warehouse system connected by elevators [18]. Werner analyzes MAPF in high-rack warehouses where agents remain on their assigned levels and elevators transport goods vertically; the method constructs waiting zones for elevator use [19]. He et al.'s CBS-E formulation models elevator capacity and reset-time conflicts [5]. These papers demonstrate that lift state and waiting near elevator shafts matter to multi-floor routing. Our evaluated contribution combines this resource with prioritized job assignment, floor-local route-leg planning, and central lift booking. The claim is limited to this implemented combination and the experiments in this paper; we make no claim of priority over all related systems.

F. Contributions

This work reimplements the ICARCV 2024 prioritized-planning baseline with corrections to the original demonstration, then evaluates a hierarchical central-plus-floor-server extension. The central component assigns jobs and books lifts, floor planners resolve local route conflicts, and lift-lobby waiting points support delays near floor transitions. An independent checker validates completed schedules. Results compare centralized, FCFS, and hybrid variants over seeded job sets, fleet sizes, job scales, and parameter settings for the tested factory layouts [2].

III. Problem Model

The model uses the prioritized assignment and route-window notation of the ICARCV baseline [1], with explicit lift and lobby elements for a multi-floor factory.

A. Sets and graph

Let $G = (N, E)$ be the undirected factory graph. Each undirected edge represents a bidirectional track. Let F be the set of floors, and let $N_f \subseteq N$ denote the nodes on floor $f \in F$. Let $L \subseteq E$ be the set of lift links between floors. The parking node is $\pi \in N$.

Let J be the set of jobs and V the set of AGVs. Each job $j \in J$ is served by exactly one AGV. Let M_i index the successive node visits on AGV i's route, including repeated visits to the same physical node.

B. Parameters

For every job $j \in J$, its pickup and delivery nodes are

$$p_j \in N, \qquad p'_j \in N, \qquad p_j \neq p'_j,$$

and t_j, t'_j are its desired pickup and delivery times. The service-time parameters are η (loading) and η' (unloading). The safety separation is $\delta > 0$, and AGV speed is $v > 0$. Each edge $e \in E$ has length ℓ_e; for lift edges $e \in L$, ℓ_e is the configured lift length. Travel time along an edge is

$$\tau_{uv} = \frac{\ell_{\{u,v\}}}{v}, \qquad \{u, v\} \in E.$$

Let $d(u, w)$ be the shortest-path travel time between nodes u, w, computed from these edge times. The code uses a fixed weight $\lambda \geq 0$ in its assignment fitness and a maximum of $k_{\max}$ passes in Algorithm 3.

Jobs are considered in the release order

$$j_1, \ldots, j_{|J|} = \mathrm{sort}_{j \in J}(t_j, j),$$

that is, increasing desired pickup time, with job identifier as the tie-breaker. AGV i has a fixed priority rank ρ_i, where smaller values mean higher priority.

C. Decision and schedule variables

The assignment variable is

$$x_{ji} \in \{0, 1\},$$

where $x_{ji} = 1$ means AGV i is assigned job j. Each job is assigned once:

$$\sum_{i \in V} x_{ji} = 1 \qquad \forall j \in J.$$

For each AGV i, let

$$R_i = (n_{i,0}, n_{i,1}, \ldots, n_{i,q_i})$$

be its ordered route, with $n_{i,0} = n_{i,q_i} = \pi$. The implemented planner constructs R_i by concatenating shortest paths from the AGV's current node to each assigned job's pickup and delivery nodes, then back to π.

For visit $m \in M_i$, let lt_{im} and rt_{im} be the start and end of its node-occupancy window. Its dwell time is $dwell_{im}$, equal to η at a pickup, η' at a delivery, and zero at a pass-through visit. The edge occupancy window for consecutive visits $m-1, m$ is

$$[rt_{i,m-1}, lt_{im}],$$

with duration fixed to the edge travel time:

$$lt_{im} - rt_{i,m-1} = \tau_{n_{i,m-1}, n_{i,m}}.$$

Thus, the AGV does not stop while traversing an edge. At ordinary nodes,

$$rt_{im} - lt_{im} = dwell_{im}.$$

The parking node is exempt from node-conflict checks and may have an extended window to represent waiting.

D. Algorithm 1 assignment quantities

When job j is considered, let s_i and vt_i be AGV i's current node and available time. The code computes

$$\begin{aligned} ct_{ji} &= vt_i + d(s_i, p_j) + \eta + d(p_j, p'_j) + \eta', \\ et_{ji} &= \max\big(0,\ t_j - \big(vt_i + d(s_i, p_j) + \eta\big)\big), \\ dt_{ji} &= \max(0,\ ct_{ji} - t'_j), \\ F_{ji} &= ct_{ji} + \lambda(et_{ji} + dt_{ji}). \end{aligned}$$

The job is assigned to the AGV minimizing F_{ji}; ties are broken by AGV priority. After assignment, vt_i and s_i are updated to ct_{ji} and p'_j, respectively.

E. Conflict separation and start delays

Let W_n be the fixed occupancy windows of higher-priority AGVs at ordinary node n. Let W'_{uv} be the fixed edge windows on directed traversal $u \to v$. In the implementation, a physical track is exclusive in both directions: edge conflicts are checked against windows for both $u \to v$ and $v \to u$.

For two different AGVs $i \neq h$, overlapping use of an ordinary node or the same bidirectional track must be separated by at least δ. For occupancy intervals $[a, b]$ and $[c, d]$, the required separation is

$$c \geq b + \delta \quad \text{or} \quad a \geq d + \delta.$$

The parking node π is exempt from node separation. Algorithm 3 resolves a conflict by shifting a prefix of the current route to the right. It computes the cumulative start delay st^* by sweeping backward from the conflict visit to the nearest previous exempt node, using the window function $f(lt, rt, W)$. After the sweep, the affected node and edge windows are shifted by st^*, and the previous parking window is extended by the same amount. The sweep is limited to $k_{\max}$ passes; if exhausted, the implementation applies its fallback delay.

F. Interpretation and assumptions

This is the implemented scheduling procedure rather than a single globally optimized mixed-integer program. It first assigns jobs greedily in pickup-release order using F_{ji}, fixes shortest-path routes, then schedules AGVs in priority order and resolves conflicts through Algorithm 3. It assumes known job times, fixed AGV priorities, constant speed, fixed edge travel times, service time only at pickup and delivery, no stops on edges, and exclusive use of each bidirectional track. The implemented centralized model allows waiting only at π; lift-lobby waiting points are an additional hybrid-layout feature.

G. Hard feasibility constraints

The following conditions express the hard checks performed by the independent verifier. Let $P_{im} = [lt_{im}, rt_{im}]$ be the window for visit m of AGV i, and let $\mathcal{B}$ be the set of exempt waiting nodes. For the centralized layout, $\mathcal{B} = \{\pi\}$; in the hybrid layout it also contains lift lobbies. Write ε for the verifier's numerical tolerance.

a) Route endpoints and initial time.: Every AGV starts at the parking node at time zero and finishes at that node:

$$n_{i,0} = \pi, \qquad n_{i,q_i} = \pi, \qquad |lt_{i,0}| \leq \varepsilon \qquad (i \in V).$$

b) Node conflict avoidance.: For each ordinary node $n \in N \setminus \mathcal{B}$, distinct AGVs may not occupy it within the safety separation. For every pair of visits (i, m) and (h, r) to n, where $i \neq h$, the windows must satisfy

$$rt_{im} + \delta \leq lt_{hr} \quad \vee \quad rt_{hr} + \delta \leq lt_{im}.$$

No node-conflict constraint is imposed on $n \in \mathcal{B}$: parking and lobbies are exempt and may hold multiple waiting AGVs.

c) Track conflict avoidance in both directions.: For each physical track $e = \{u, v\} \in E$, define the traversal window for a visit transition $m - 1 \to m$ by

$$Q_{im} = [a_{im}, b_{im}] = [rt_{i,m-1}, lt_{im}], \qquad \{n_{i,m-1}, n_{i,m}\} = e.$$

For every pair of traversals of the same physical track by distinct AGVs, including traversals in opposite directions, require

$$b_{im} + \delta \leq a_{hr} \quad \vee \quad b_{hr} + \delta \leq a_{im}.$$

This is the verifier's bidirectional-track rule: use of $u \to v$ conflicts with both $u \to v$ and $v \to u$ use by another AGV.

d) No pause on an edge.: Every edge window has exactly the edge's non-stop travel duration:

$$lt_{im} - rt_{i,m-1} = \tau_{n_{i,m-1}, n_{i,m}} = \frac{\ell_{\{n_{i,m-1}, n_{i,m}\}}}{v} \qquad (m = 1, \ldots, q_i).$$

e) Dwell times.: Let $\mathcal{P}_{im}$ and $\mathcal{D}_{im}$ indicate that visit (i, m) performs a pickup or delivery, respectively. With at most one service event per visit, define

$$d_{im} = \eta\, \mathcal{P}_{im} + \eta'\, \mathcal{D}_{im}.$$

At every non-exempt node, dwell must include the required service time (and is zero at pass-through visits in the constructed plans):

$$rt_{im} - lt_{im} = d_{im} \qquad (n_{i,m} \notin \mathcal{B}).$$

At an exempt parking or lobby visit, the verifier only requires the window to be at least as long as any service dwell there:

$$rt_{im} - lt_{im} \geq d_{im} \qquad (n_{i,m} \in \mathcal{B}),$$

so the excess duration can represent waiting. In the implemented job routes, η is used only at pickup events, η' only at delivery events, and pass-through visits have zero dwell.

f) Exactly-once job service.: Let z_{ij} be the number of times job j appears in AGV i's assigned-job list. Every job must be assigned to exactly one AGV:

$$\sum_{i \in V} z_{ij} = 1 \qquad (j \in J), \qquad z_{ij} \in \mathbb{Z}_{\geq 0}.$$

Equivalently, the multiset union of all AGV job lists must equal the job set, with each job occurring once. The verifier also requires exactly one pickup and one delivery event for each job, at the specified stations and on the assigned AGV, with pickup preceding delivery and pickup completion preceding delivery completion.

H. Algorithm 3 start-delay conditions

For a conflict at visit p, Algorithm 3 chooses a nonnegative cumulative right-shift $st^* \geq 0$ for the route segment after the nearest preceding exempt node q through p. In the no-fallback case, its update is

$$lt'_{im} = lt_{im} + st^*, \qquad rt'_{im} = rt_{im} + st^* \quad (q < m \leq p), \qquad rt'_{iq} = rt_{iq} + st^*.$$

The last equality extends the waiting window at the exempt node. Edge windows within the shifted segment move consistently because both endpoint times are shifted; the connecting edge from q is delayed by extending rt_{iq}. The algorithm accumulates the smallest right-shifts needed to clear the fixed higher-priority windows encountered in its backward sweep, while checking that a feasible shift does not cross the next blocking window (the mt^* bound). It repeats up to $k_{\max}$ passes. If those passes are exhausted, the code uses its fallback shift so the AGV waits at the previous exempt node until higher-priority traffic is free. The independent verifier subsequently checks the resulting complete plan.

I. Reported objectives

The implementation reports three performance measures. Let C_j be the actual delivery-completion time of job j in the final plan, and let U_i be AGV i's usage time, defined in the code as its return time to π when it serves jobs and zero otherwise. Then

$$\begin{aligned} M &= \max_{j \in J} C_j && \text{(makespan, last delivery)}, \\ T &= \sum_{j \in J} \max(0, C_j - t'_j) && \text{(total delivery tardiness)}, \\ I &= \max_{i \in V} U_i - \min_{i \in V} U_i && \text{(load imbalance)}. \end{aligned}$$

One may state a scalar objective as

$$\min \ \alpha M + \beta T + \gamma I, \qquad \alpha, \beta, \gamma \geq 0,$$

or a lexicographic objective as

$$\text{lexmin}(M, T, I).$$

These are possible mathematical formulations for comparing schedules; the code does not choose weights or optimize either combined objective. It uses F_{ji} greedily for job assignment, then applies prioritized conflict resolution, and reports M, T, and I as metrics.

a) Enforced versus checked conditions.: Algorithm 2 constructs each route's node and edge windows and calls Algorithm 3 when a new window conflicts with already frozen higher-priority windows. Algorithm 3 shifts the affected route prefix, extends the preceding exempt node's window, and thereby enforces conflict avoidance incrementally; in the hybrid planner, floor servers also plan legs and the central server reserves lift tracks. The independent verifier does not plan or repair the schedule. It checks the completed plans for start/end at π, route continuity, node and bidirectional-track separation, lift capacity, lobby capacity, exact edge travel durations, service dwell, and exactly-once pickup and delivery events at the required stations and in the required order. Thus, the full-plan conditions are independently checked after Algorithms 2 and 3 construct the schedule.

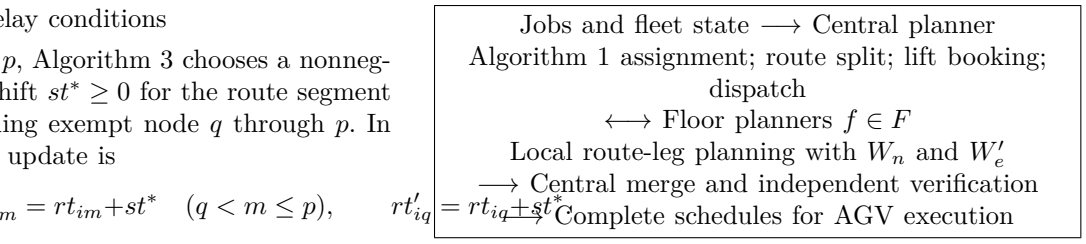


Fig. 1. Central coordination and floor-local route planning.

IV. System Architecture and Methods

The method combines the prioritized assignment and route-window procedures of the centralized baseline [1] with a hierarchical floor decomposition. The central component assigns jobs, splits routes at lifts, arbitrates lift use, and merges the resulting schedules. A floor-specific planner constructs conflict-free timings for its local route legs.

A. Architecture classification

The architecture is hierarchical and decomposed, with floor-local planning components; it is not fully decentralized. Job assignment, lift booking, route-leg dispatch, final-plan merging, and verification remain centrally orchestrated. Floor planners maintain local node and track reservation sets W_n and W'_e, and resolve conflicts on their own route legs. The central lift calendar coordinates the cross-floor resource.

Figure 1 summarizes the information flow. The central-to-floor exchanges are internal planner requests and reservation responses in the evaluated implementation. The AGVs do not negotiate jobs or lift slots directly.

B. Algorithm 1: prioritized job assignment

Jobs are processed in increasing desired pickup time, with job identifier as the tie-breaker. For each candidate AGV i, the planner evaluates $F_{ji} = ct_{ji} + \lambda(et_{ji} + dt_{ji})$ using predicted completion time, pickup earliness, and delivery tardiness. It assigns the job to the AGV with minimum fitness; ties use fixed AGV priority. The selected AGV's route is extended through shortest paths to pickup and delivery, with loading and unloading dwell times η and η', and is ultimately extended back to parking π.

C. Algorithm 2: prioritized conflict-free routing

Algorithm 2 processes AGVs in ascending fixed priority, $V_1 \to V_{|V|}$. For each route, it constructs node occupancy windows $[lt_m, rt_m]$ and edge traversal windows. If a window conflicts with a frozen reservation, Algorithm 3 delays the affected route segment. The resulting node and

Algorithm 1 Prioritized job assignment

Require: Jobs J, AGVs V, fitness weight λ, service times η, η'
Ensure: Assignment x_{ji} and complete routes R_i

for each $j \in J$ in order (t_j, j) do
  for each $i \in V$ do
    Compute $ct_{ji}, et_{ji}, dt_{ji}$ from AGV state (s_i, vt_i).
    $F_{ji} \leftarrow ct_{ji} + \lambda(et_{ji} + dt_{ji})$.
  end for
  Choose $i^* \leftarrow \arg\min_i(F_{ji}, \rho_i)$; set $x_{ji^*} \leftarrow 1$.
  Append shortest paths $s_{i^*} \rightarrow p_j \rightarrow p'_j$ with dwell η, η'.
  Update AGV state to (p'_j, ct_{ji^*}).
end for
for each $i \in V$ do
  Append a shortest path from the final assigned station to π.
end for

Algorithm 2 Prioritized conflict-free routing

Require: Routes R_i, dwell times, priority order $V_1, \ldots, V_{|V|}$
Ensure: Timed plans and frozen reservations W_n, W'_e

Initialize empty node and edge reservation sets W_n, W'_e.
for each AGV i in priority order do
  Construct node and edge windows along R_i.
  for each candidate window conflicting with W_n or W'_e do
    Apply Algorithm 3 to delay the route segment.
  end for
  Freeze ordinary node windows in W_n and track windows in W'_e.
end for

track windows are then added to the frozen sets W_n and W'_e. In the paper-style configuration, only parking π is an exempt waiting node; the lobby configurations also exempt lift-lobby nodes.

D. Algorithm 3: sliding-window conflict elimination

For a candidate interval $[lt, rt]$ and fixed windows W, the interval function $f(lt, rt, W)$ returns the smallest right shift st that clears the candidate with safety separation δ, together with the available shift bound mt before the next blocking window. For a conflict at route visit p, the algorithm sweeps backward to the nearest preceding exempt node q, accumulating the route-segment shift st^* and maintaining the bound mt^*. It repeats the sweep up to $k_{\max}$ passes. On exhaustion, the fallback delays departure at the exempt node until the blocking traffic is free. The shifted windows from $q+1$ through p move by st^*, and the window at q is extended by that delay.

Algorithm 3 Sliding-window conflict elimination

Require: Conflict visit p, reservations W, route R, limit $k_{\max}$
Ensure: Conflict-free shifted windows and delay st^*

Find the nearest preceding exempt visit q.
for $k = 1$ to $k_{\max}$ do
  Set $st^* \leftarrow 0$ and $mt^* \leftarrow \infty$.
  for visits $m = p, p-1, \ldots, q+1$ do
    Evaluate $f(lt_m + st^*, rt_m + st^*, W)$; update mt^* and st^*.
    if $st^* > mt^*$ then
      Stop this backward sweep.
    end if
  end for
  if the sweep completes within mt^* then
    break
  end if
end for
if no feasible sweep completed then
  Apply the fallback delay at q until blocking reservations clear.
end if
Shift windows $q+1, \ldots, p$ and extend the window at q by st^*.

E. Hybrid central-plus-floor-server extension

The hybrid planner cuts each complete route at lift edges into floor-local legs. Each floor planner schedules a leg using local W_n and W'_e; when needed, the AGV waits at its entry lobby or at π. The central planner reserves the lift track at the earliest available time after the AGV is ready, respecting the safety separation. It then releases the next floor leg at the arrival lobby and merges floor and lift windows into the complete plan before verification.

Legs are dispatched using either priority ordering or first-come-first-served (FCFS). FCFS orders legs by release time, AGV priority, then leg index. The priority order processes the legs of higher-priority AGVs first and is used as a decomposition check. In the tested seeds, priority-order hybrid plans match the centralized-with-lobbies plans. For the principal performance comparison, configuration C uses FCFS order.

V. Experimental Setup

The factory graphs, job sets, and planner runs are generated by the project implementation [2].

A. Factory layout

Experiments use a three-floor factory layout. It contains three floor graphs with nine operational nodes per floor, for 27 operational nodes total, connected by 38 edges. Four inter-floor edges represent lifts; each lift has a configured length of 40 m. The fleet's single parking node is $\pi = 1\text{A}$. For configurations with lobbies, an additional lobby node is inserted at each lift landing. Lobby nodes are waiting

Algorithm 4 Hybrid floor-leg planning and lift booking

Require: Assigned routes, floor planners, central lift calendar

Ensure: Merged complete AGV plans

1: Split each route into maximal floor-local legs at every lift.
2: Insert each initial leg into the dispatch queue using the chosen order.
3: while the queue is nonempty do
4: Remove the next leg and schedule it with its floor planner.
5: if the route continues on another floor then
6: Reserve the intervening lift trip centrally.
7: Release the next leg at the arrival lobby after lift travel.
8: end if
9: end while
10: Merge floor-leg and lift windows; independently verify each plan.

locations and are excluded from the job-station sampling set. Thus, jobs are sampled from the 26 non-parking, non-lobby operational nodes.

B. Jobs and fleet

The job generator creates each job by sampling two distinct stations uniformly from the 26 operational nodes: pickup p_j and delivery p'_j. Desired pickup time is drawn uniformly over the 600-second horizon,

$$t_j \sim U[0, 600].$$

For each job, an independent slack value is drawn as $s_j \sim U[30, 200]$ and the desired delivery time is set to

$$t'_j = t_j + d(p_j, p'_j) + \eta + \eta' + s_j,$$

where $d(p_j, p'_j)$ is the shortest-path travel time on the factory graph. The implementation rounds generated times to one decimal place. The random seed determines the job set; the comparison uses seeds 1 through 10.

The main experiment uses 20 vehicles named AGV-01 through AGV-20, with fixed priority ranks V_1 through V_{20} in that order. The fleet-size sweep repeats the experiment with 4, 8, 12, 16, and 20 AGVs. Unless otherwise specified, the main comparison uses 100 jobs and 20 AGVs per seed.

C. Planning parameters and configurations

All configurations use loading time $\eta = 16$ s, unloading time $\eta' = 16$ s, safety separation $\delta = 10$ s, fitness weight $\lambda = 1.0$, and vehicle speed $v = 1$ m/s. Lift links use their configured 40 m length, so their non-stop travel time is 40 s at the specified speed.

The comparison evaluates four configurations on the same generated job sets:

A — Centralized prioritized planner with waiting permitted at the parking node only; lift landings have no exempt waiting lobbies.

B — Centralized with lobbies. The centralized planner uses the same route and conflict-planning procedure, with lift-lobby nodes added as exempt waiting locations.

B2 — Centralized FCFS legs. A single centralized planner holds all reservations and processes route legs in the same FCFS order as C.

C — Hybrid. The hybrid central-plus-floor-server planner uses the same lobbies. The central server assigns jobs and books lift trips; floor servers plan floor legs in first-come-first-served order (order="fcfs").

The main comparison contains 10 seeds (1–10), 100 jobs, and 20 AGVs for each of A, B, B2, and C. A separate fleet sweep evaluates 4, 8, 12, 16, and 20 AGVs over the same seeds and job-generation settings. For correctness checking of the decomposition, the hybrid planner can also be run in priority order; that check compares its plans with configuration B.

Additional studies use the same job and fleet model unless otherwise noted. The job-scale sweep fixes 20 AGVs, uses 5 seeds, and tests 50, 100, 200, and 400 jobs for A, B, B2, C, and prioritized wait adjustment. The five-floor sensitivity study uses 50 jobs, 10 AGVs, 3 seeds, and all combinations of $\eta \in \{8, 16, 32\}$ s, $\delta \in \{5, 10, 20\}$ s, and lift length in $\{20, 40, 80\}$ m. The assignment-baseline comparison uses 100 jobs, 20 AGVs, and 3 seeds; the competing wait-adjustment comparison uses 10 seeds. Lobby capacity and wait-fairness studies use 100 jobs and 20 AGVs over 10 seeds.

D. Evaluation metrics

The comparison records the following measures:

- Makespan: time at which the last job delivery is completed.
- Tardiness: total positive delivery lateness across jobs, $\sum_j \max(0, C_j - t'_j)$, where C_j is actual delivery completion.
- Load imbalance: maximum minus minimum AGV usage time, where usage time is measured until return to parking (zero for an AGV serving no jobs).
- Waiting: total and maximum per-AGV waiting time at parking or lift lobbies.
- Compute time: total planning CPU time across servers and CPU time at the busiest server; wall-clock planning time is also recorded.
- Server state: maximum number of reservation windows held by any one server.
- Communication: central-to-floor message count and the number of window records in the transmitted payload.
- Conflict-resolution effort: number of Algorithm 3 calls.

The comparison reports means over the ten seeds; the figures also show the seed range where applicable. Fleet-sweep results report the corresponding metrics by fleet size.

E. Plan verification

Every generated schedule is passed to an independent verifier. It evaluates the completed plans for start and end at π, node conflict separation (with parking/lobbies exempt), bidirectional track conflict separation, no pause during edge traversal, correct node-window dwell duration, and exactly-once job service. For the main A/B/C comparison, a run is accepted only when the verifier returns no violations. Configuration C in priority order is additionally compared against B to check that the server decomposition preserves the centralized-with-lobbies plans on the tested seeds.

F. Software and execution environment

The software environment uses Python 3.11 or 3.12 and the dependencies networkx, matplotlib, numpy, and pillow. The dependency manifest names these packages but does not pin their individual versions. The repository does not record processor, memory, operating-system, or accelerator specifications; therefore, hardware details and exact installed package versions are not reported here. Runtime measurements should be interpreted in light of this missing machine-level metadata.

VI. Results and Discussion

Unless stated otherwise, results below are generated by the project implementation [2]. The principal comparison uses 100 jobs and 20 AGVs on each of ten job sets (seeds 1–10). A is the centralized paper-style planner, B is centralized with lobbies, B2 is centralized with lobbies and FCFS leg order, and C is the hybrid with FCFS leg order. Every schedule contributing a reported metric in the principal comparison, job-scale, sensitivity, assignment-baseline, competing-planner, and fairness experiments passed agv_core.verify(). The lobby-capacity sweep also contains B runs that failed verification; those failures and verified-run counts are reported explicitly below.

A. Correctness and decomposition check

All A, B, B2, and C plans in the principal comparison pass verification. The B2 and C schedules match on all ten seeds, and the priority-order hybrid reproduces B on all ten seeds. In these experiments, the floor-server split changes reservation state and computation distribution but not the schedule metrics when the route-leg order is held fixed. Figure 4 illustrates the resulting hybrid route of AGV-03 on seed 7: floor legs joined by lift trips, with the waiting between legs placed at the lift lobbies.

B. Mean performance over ten seeds

Table I reports the ten-seed means. The percentage column is C relative to A. In this tested three-floor setup, C reduces makespan by 65%, tardiness by 72%, and imbalance by 80%. Its busiest-server CPU mean is 7.6 ms versus 92.5 ms for A (92% lower); total CPU across C's servers is 21.9 ms. Planning CPU is wall-clock dependent: across five repeated runs of this experiment (seeds 1–10, outputs in results_cpu_repeat_r1–r5), the busiest-server reduction stays between 92% and 93% (medians 98.6 ms for A and 7.4 ms for C; full ranges are recorded in results_manifest.json). B2 and C have equal scheduling outcomes, so the B2-to-C contrast isolates the floor server split's measured computation/state distribution rather than a schedule-quality change.

Figure 2 visualizes the ten-seed means and ranges. Figure 3 shows the A and C plans for the seed-7 job set, and Figure 4 shows how C cuts one AGV's route into floor legs joined by lift trips.

C. Fleet-size sweep

Table II gives ten-seed mean imbalance and tardiness. Tardiness decreases with fleet size for A at every tested step. For B it decreases through 16 AGVs, then rises from 117,682 s at 16 AGVs to 122,279 s at 20. For B2 and C it decreases slightly from 90,343 s to 90,104 s over that final step. Thus tardiness does not fall monotonically in all configurations. Imbalance falls for A and rises for B, B2, and C over this sweep. B2 and C have the same values at every tested fleet size.

Figure 5 shows all three fleet-sweep metrics.

D. Job-scale sweep

The job-scale experiment fixes 20 AGVs and evaluates 50, 100, 200, and 400 jobs over seeds 1–5. Table III reports C's means and its percentage reductions relative to A. B2 and C have identical outcomes at each tested scale. The C-to-A reduction grows across these four tested workload sizes for all three metrics. This is an observed trend for the tested layout, fleet, and seeds, not a general scaling guarantee.

E. Parameter sensitivity on five floors

The sensitivity run covers a five-floor layout, 50 jobs, 10 AGVs, and three seeds for each of 27 parameter triples: $\eta \in \{8, 16, 32\}$ s, $\delta \in \{5, 10, 20\}$ s, and lift length in $\{20, 40, 80\}$ m. All 324configuration rows pass verification. Table IV gives the C means at each level of one factor, averaged over the other two factors and the three seeds. The outcomes generally rise with larger δ and lift length in these marginal means; the imbalance mean does not rise monotonically with η. Because this is a finite grid of settings on one layout, it does not establish behavior outside that grid.

F. Assignment baselines

We regenerated the FCFS and random assignment baselines using the common Algorithm 2/3 routing planner, for 100 jobs and 20 AGVs over seeds 1–3. All five configurations passed verification on each seed. Table V reports means from the per-seed baseline output. Under this planner and these three seeds, Algorithm 1's fitness-based assignment gives little benefit over the FCFS assignment baseline: FCFS has slightly lower mean makespan,

TABLE I
Mean metrics over ten seeds (mean ± sample standard deviation over seeds). Percentages compare C with A.

Metric	A	B	B2	C	C vs. A
Last-delivery makespan (s)	$7,972.4 \pm 819.1$	$3,705.9 \pm 309.8$	$2,768.9 \pm 195.8$	$2,768.9 \pm 195.8$	−65%
Total tardiness (s)	$316,893.6 \pm 41,316.3$	$122,278.8 \pm 11,430.7$	$90,104.5 \pm 11,447.3$	$90,104.5 \pm 11,447.3$	−72%
Load imbalance (s)	$7,149.9 \pm 772.6$	$2,832.3 \pm 291.4$	$1,441.2 \pm 241.1$	$1,441.2 \pm 241.1$	−80%
Total waiting (s)	$64,689.7 \pm 8,466.2$	$27,647.7 \pm 2,747.6$	$27,104.5 \pm 2,180.2$	$27,104.5 \pm 2,180.2$	−58%
Longest AGV wait (s)	$7,133.4 \pm 817.9$	$2,830.8 \pm 302.2$	$1,887.7 \pm 173.9$	$1,887.7 \pm 173.9$	−74%
Busiest-server CPU (ms)	92.5 ± 23.0	19.9 ± 8.7	22.9 ± 2.7	7.6 ± 1.2	−92%
Total server CPU (ms)	92.5 ± 23.0	19.9 ± 8.7	22.9 ± 2.7	21.9 ± 2.0	−76%
Windows on busiest server	$2,030.7 \pm 84.5$	$1,733.5 \pm 69.5$	$1,733.5 \pm 69.5$	634.5 ± 30.0	−69%
Central–floor messages	0.0 ± 0.0	0.0 ± 0.0	337.2 ± 16.2	337.2 ± 16.2	—
Algorithm 3 calls	106.6 ± 7.6	165.1 ± 12.1	168.6 ± 16.4	168.6 ± 16.4	+58%

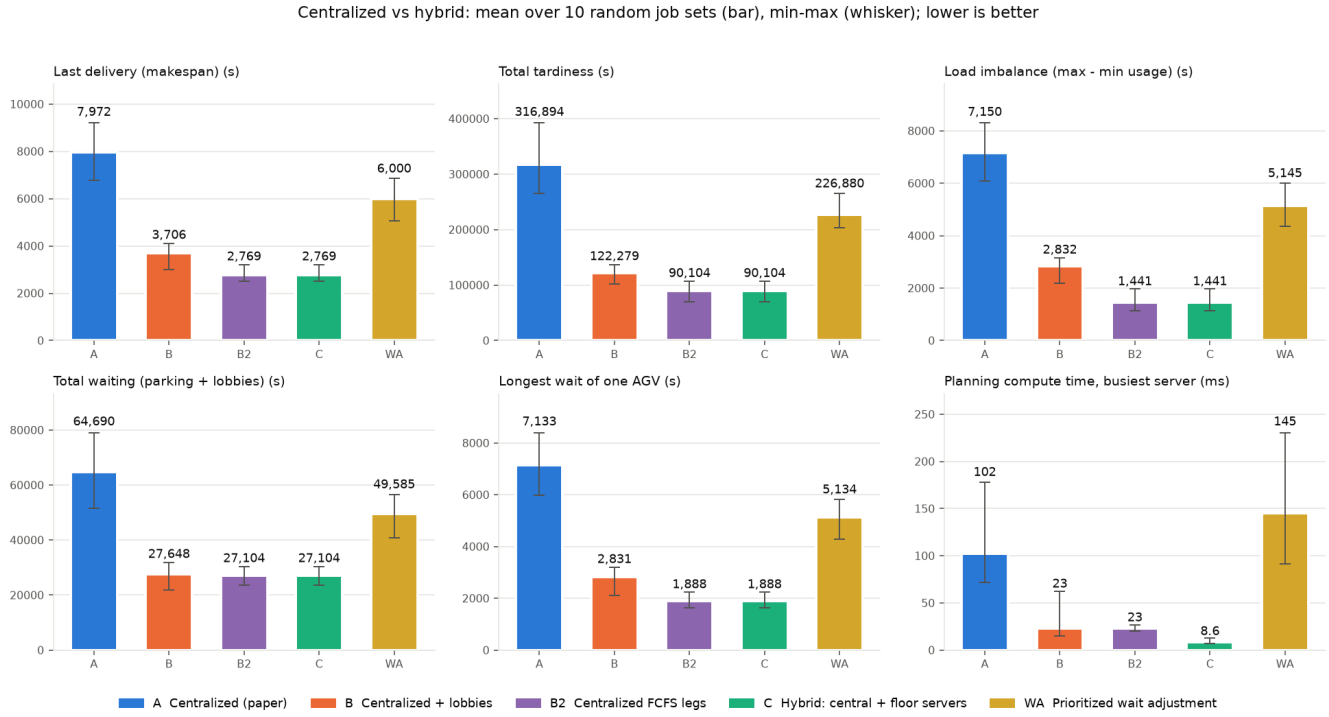


Fig. 2. Mean and min–max range over the ten job sets for A, B, B2, and C.

tardiness, and imbalance. Random assignment is substantially worse on all three measures. This small three-seed comparison does not establish a general ranking.

G. Competing prioritized wait-adjustment planner

The competing planner (WA) computes individual paths and inserts waits to avoid conflicts using the same seeded jobs, layout, fleet, speeds, and safety gap. In the 10-seed, 100-job, 20-AGV comparison, every WA plan passed verification. Its mean makespan, tardiness, and imbalance were 5,999.9 s, 226,879.9 s, and 5,144.8 s, respectively, compared with 2,768.9 s, 90,104.5 s, and 1,441.2 s for C. The job-scale sweep also reports higher WA means than C at 50, 100, 200, and 400 jobs. On these tested cases WA did not outperform the hybrid; this is not evidence that it cannot do so on other instances.

H. Finite lobby capacity

The capacity experiment evaluates B and C on three- and five-floor layouts, 100 jobs, 20 AGVs, and ten seeds. Table VI reports B's verified-plan counts and means when available, and C's verified counts and means. The summary describes finite-capacity C as using conservative FCFS tour admission, with other AGVs waiting at π to avoid lobby wait-for cycles. Its finite-capacity plans verify, but the resulting schedules are much slower than the unlimited-capacity case. Some B seeds failed plan verification; that is a planner failure in those runs, not proof that the underlying scheduling instances are physically infeasible.

I. Per-server state and computation

For seed 7, the centralized planner reports 1,989 windows and 92.8 ms CPU on A's central server. C holds

TABLE II
Fleet sweep: mean ± sd load imbalance and total tardiness (s), ten seeds.

AGVs	Metric	A	B	B2	C
4	imbalance (s)	$9,119 \pm 1,420$	$2,043 \pm 377$	518 ± 219	518 ± 219
4	tardiness (s)	$602,120 \pm 102,189$	$256,490 \pm 21,221$	$235,718 \pm 14,871$	$235,718 \pm 14,871$
8	imbalance (s)	$8,394 \pm 1,186$	$2,346 \pm 334$	788 ± 201	788 ± 201
8	tardiness (s)	$453,005 \pm 58,101$	$149,864 \pm 14,928$	$126,223 \pm 12,444$	$126,223 \pm 12,444$
12	imbalance (s)	$8,249 \pm 763$	$2,456 \pm 358$	$1,145 \pm 256$	$1,145 \pm 256$
12	tardiness (s)	$421,464 \pm 50,500$	$125,292 \pm 21,328$	$100,319 \pm 13,957$	$100,319 \pm 13,957$
16	imbalance (s)	$7,466 \pm 870$	$2,715 \pm 316$	$1,230 \pm 161$	$1,230 \pm 161$
16	tardiness (s)	$348,159 \pm 52,200$	$117,682 \pm 10,762$	$90,343 \pm 9,685$	$90,343 \pm 9,685$
20	imbalance (s)	$7,150 \pm 773$	$2,832 \pm 291$	$1,441 \pm 241$	$1,441 \pm 241$
20	tardiness (s)	$316,894 \pm 41,316$	$122,279 \pm 11,431$	$90,104 \pm 11,447$	$90,104 \pm 11,447$

TABLE III
Job-scale means: C metrics (mean ± sd, seeds 1–5) and C reduction relative to A.

Jobs	C makespan (s)	C tardiness (s)	C imbalance (s)	Makespan reduction	Tardiness reduction	Imbalance reduction
50	$1,895.9 \pm 307.7$	$22,311.8 \pm 5,964.3$	$1,425.6 \pm 572.8$	58.0%	68.3%	66.0%
100	$2,725.3 \pm 137.1$	$83,285.2 \pm 9,346.9$	$1,430.3 \pm 141.2$	65.2%	72.9%	79.6%
200	$4,674.2 \pm 142.1$	$360,814.5 \pm 6,079.8$	$1,576.1 \pm 169.0$	72.6%	76.6%	89.8%
400	$8,541.3 \pm 345.2$	$1,484,910.4 \pm 50,634.9$	$2,350.0 \pm 437.8$	77.5%	78.9%	93.3%

TABLE IV
Hybrid C marginal means by parameter value (mean ± sd) in the five-floor sweep.

Factor	Value	Makespan (s)	Tardiness (s)	Imbalance (s)
η (s)	8	$2,377.0 \pm 945.2$	$36,979.1 \pm 24,312.8$	845.3 ± 313.7
η (s)	16	$2,431.2 \pm 882.6$	$38,716.9 \pm 24,304.4$	892.2 ± 374.0
η (s)	32	$2,668.6 \pm 926.8$	$42,874.4 \pm 24,063.4$	811.6 ± 252.5
δ (s)	5	$2,290.2 \pm 859.0$	$34,629.3 \pm 23,121.4$	754.9 ± 290.3
δ (s)	10	$2,457.9 \pm 904.7$	$38,250.9 \pm 23,793.8$	833.5 ± 295.0
δ (s)	20	$2,728.8 \pm 960.9$	$45,690.1 \pm 24,761.6$	960.8 ± 334.3
Lift length (m)	20	$1,689.3 \pm 338.8$	$19,697.5 \pm 8,465.4$	557.1 ± 153.3
Lift length (m)	40	$2,307.7 \pm 485.5$	$33,839.1 \pm 12,321.8$	797.2 ± 169.6
Lift length (m)	80	$3,479.9 \pm 716.4$	$65,033.7 \pm 20,949.7$	$1,194.9 \pm 194.1$

TABLE V
Assignment baselines: means ± sd over seeds 1–3 (100 jobs, 20 AGVs).

Configuration	Makespan (s)	Tardiness (s)	Imbalance (s)
A: Algorithm 1 assignment	$7,888.6 \pm 1,134.1$	$316,407.8 \pm 65,925.4$	$7,048.3 \pm 1,081.2$
B: centralized with lobbies	$3,687.0 \pm 159.4$	$120,550.8 \pm 14,279.8$	$2,858.9 \pm 164.4$
C: hybrid	$2,783.2 \pm 101.8$	$84,528.3 \pm 12,996.4$	$1,431.9 \pm 199.1$
FCFS assignment	$7,750.5 \pm 1,267.6$	$304,553.0 \pm 76,618.6$	$6,971.3 \pm 1,150.7$
Random assignment	$12,322.1 \pm 1,520.5$	$493,263.8 \pm 86,185.2$	$11,959.8 \pm 1,744.8$

TABLE VI
Finite-lobby results: verified runs out of ten and mean ± sd makespan / tardiness (s).

Layout	K	B verif.	B makespan (s)	B tardiness (s)	C verif.	C makespan (s)	C tardiness (s)
3 floor	1	0/10	n/a	n/a	10/10	$19,153.1 \pm 649.9$	$911,565.3 \pm 35,190.0$
3 floor	2	2/10	$4,108.9 \pm 188.5$	$146,026.1 \pm 2,430.2$	10/10	$19,153.1 \pm 649.9$	$911,565.3 \pm 35,190.0$
3 floor	4	10/10	$3,764.2 \pm 374.8$	$124,813.4 \pm 12,984.3$	10/10	$19,153.1 \pm 649.9$	$911,565.3 \pm 35,190.0$
3 floor	Unlimited	10/10	$3,705.9 \pm 309.8$	$122,278.8 \pm 11,430.7$	10/10	$2,768.9 \pm 195.8$	$90,104.5 \pm 11,447.3$
5 floor	1	0/10	n/a	n/a	10/10	$26,701.3 \pm 1,164.7$	$1,294,641.3 \pm 72,645.6$
5 floor	2	0/10	n/a	n/a	10/10	$26,701.3 \pm 1,164.7$	$1,294,641.3 \pm 72,645.6$
5 floor	4	9/10	$4,059.7 \pm 318.2$	$134,817.4 \pm 19,342.2$	10/10	$26,701.3 \pm 1,164.7$	$1,294,641.3 \pm 72,645.6$
5 floor	Unlimited	10/10	$4,014.8 \pm 339.4$	$133,751.7 \pm 19,177.3$	10/10	$3,174.2 \pm 299.1$	$110,421.4 \pm 14,112.9$



Fig. 3. How to read: each row is one AGV in priority order (V01 has the highest priority), colors give the activity (moving on floor 1, 2, or 3, lift trip, loading η, unloading η'), and hatched bars are waiting. The panels show the seed-7 job set under A (top) and C (bottom) on a shared time axis; dashed lines mark each panel's last delivery. What to notice: under A the rows form a staircase of waiting that descends with priority. The annotation marks AGV-20: it waits 6,737.3 s before its first move under A but only 375.0 s under C, while its service time is 160.0 s in both configurations. Panel titles give the data-derived totals: last delivery 7,547.5 versus 2,549.8 s, load imbalance 6,734.6 versus 1,503.8 s, and total waiting 62,169.9 versus 26,293.1 s.

146 windows and uses 7.7 ms on its central server; its floor servers hold 478, 619, and 454windows and use 3.6, 7.4, and 5.2 ms, respectively. The largest C server therefore holds 619 windows and uses 7.4 ms in this seed. These seed-specific values are distinct from the ten-seed means in Table I. The server state and CPU comparison appears in Figure 6.

J. Discussion and scope

In the tested three-floor setting, adding lobbies improves the A-to-B comparison. Changing B to B2 changes route-leg processing to FCFS and improves the reported scheduling outcomes. B2 and C then match schedule metrics, so the additional floor-server split changes measured state and computation distribution, not schedule quality, in these comparisons. The five-floor sensitivity and lobby-capacity experiments show that outcomes depend on parameter values and on the capacity-handling policy. Finite capacity under the tested conservative admission rule can impose large delays. The job-scale sweep indicates a larger C-versus-A reduction over the four tested workloads, while the fleet sweep has the non-monotone B tardiness result described above. None of these finite experiments establishes performance outside the layouts, seeds, fleet sizes, and parameters tested.

VII. AGV Waiting-Time Fairness

We measured each vehicle's total wait at parking nodes and lift lobbies for configurations A, B, B2, and C over seeds 1–10 (100 jobs and 20 AGVs per seed). Each schedule was checked by the independent verifier; all plans passed [2]. The table reports the mean and standard deviation across the ten per-seed fairness statistics. Pearson and Spearman correlations use the fixed AGV priority rank (1 is highest); Gini is calculated over per-AGV total wait. The max/min ratio is infinite whenever any AGV has zero wait.

The figure plots mean total wait by fixed priority rank. A and B concentrate delay strongly toward lower-priority ranks. B2 and C have identical wait statistics in this experiment, consistent with their identical FCFS leg ordering and schedules: the floor-server split itself did not change fairness here. C reduces inequality relative to A and B (lower Gini and a smaller priority-wait correlation),

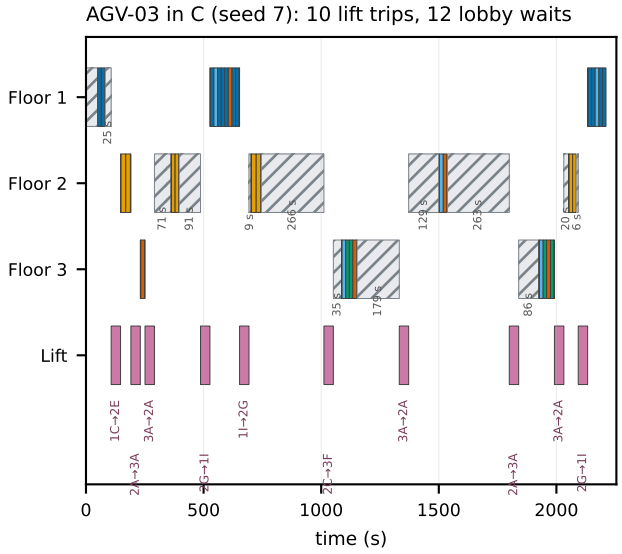


Fig. 4. How the hybrid builds one route, shown for AGV-03 under C on the same seed. Rows are the three floors and the lifts; the AGV's route is cut into floor legs joined by 10 lift trips (labelled with their lobby-to-lobby nodes). Hatched bars at the lift-entry lobbies are lobby waits: 12 waits totalling 1,179.5 s. Notice the repeated pattern travel, lobby wait, lift trip: between floor legs the AGV waits at lift lobbies rather than holding a floor node.

TABLE VII
Waiting fairness by configuration, mean ± standard deviation over ten seeds.

Configuration	Pearson r	Spearman ρ	Gini	Max/min wait
A	0.884 ± 0.073	0.886 ± 0.074	0.382 ± 0.018	∞ (10/10)
B	0.961 ± 0.041	0.965 ± 0.033	0.354 ± 0.016	∞ (10/10)
B2	0.419 ± 0.267	0.425 ± 0.264	0.143 ± 0.036	3.842 ± 1.468
C	0.419 ± 0.267	0.425 ± 0.264	0.143 ± 0.036	3.842 ± 1.468

but the top-ranked AGV waits longer on average in C (1,169.1 s) than under A or B (0.0 s in each). Thus the hybrid is fairer by these aggregate distribution and rank-correlation measures, while shifting some delay onto the highest-priority vehicle; this is not an unqualified fairness improvement.

A. Fairness variants on C

Two simple changes were evaluated on the same ten seeds. In the rotation variant, job batches rotate the AGV priority ordering. In the waiting-trigger variant, AGVs whose accumulated wait exceeded 600 s were promoted during replanning; 600 s is a trigger, not a hard cap. All variant plans passed the independent verifier.

Rotation lowered the reported mean makespan, tardiness and imbalance slightly, but the Gini increased from 0.143 to 0.145; it weakened correlation with the original fixed ranks but did not reduce overall wait inequality. The wait trigger produced the same schedules and performance metrics as baseline C: under the current FCFS leg order, priority changes mostly affect ties and did not alter route-leg order. These results are descriptive for ten seeds; no significance test was performed [2].

TABLE VIII
C fairness variants: mean ± standard deviation over ten seeds.

Variant	Makespan (s)	Tardiness (s)	Imba
C (baseline)	2768.9 ± 195.8	90104.5 ± 11447.3	1441.2
C with rotating priority	2737.6 ± 169.7	88009.6 ± 10147.2	1425.3
C with 600 s wait-trigger boost	2768.9 ± 195.8	90104.5 ± 11447.3	1441.2

VIII. Limitations

The results should be interpreted in light of the implementation's scope and modeling assumptions [2]. Some are deliberate simplifications that make the planning procedure testable; others are limitations of the current prototype that constrain its transfer to operational factories.

First, this is a simulation implementation. It has no robot operating system (ROS) integration, vehicle localization or control interface, industrial communications layer, or hardware-in-the-loop evaluation. This is a known implementation limitation: simulation verification does not establish safe execution under sensor error, vehicle dynamics, communication delay, or controller behavior.

Second, each scenario supplies a static set of jobs known before planning. The planner sorts jobs by desired pickup time and does not handle new arrivals during execution. This is a deliberate batch-scheduling assumption, but it limits conclusions about online manufacturing where urgent jobs, cancellations, or changed deadlines require replanning. The fleet priority order is also fixed (AGV-01 as V_1 through AGV-20 as V_{20}); priorities are not updated to reflect waiting time, task urgency, or changing vehicle state. This deliberate rule simplifies deterministic comparisons, while potentially creating unfair delays and making performance dependent on the chosen order.

Third, every generated job is assigned and served exactly once. There is no job reassignment, task preemption, vehicle breakdown model, or recovery procedure after a failure. Exactly-once service is part of the modeled problem, whereas the absence of failure handling is a known limitation for deployment in factories where vehicles and stations may become unavailable.

Fourth, the parking node π is deliberately exempt from node-conflict checks so multiple AGVs may start and wait there. In the hybrid configuration, lift lobbies can also be exempt waiting locations; lobby construction is optional. The model does not impose a finite capacity on these exempt waiting areas. Real parking bays and lift lobbies may have limited space, so the assumption can understate congestion and should be revisited for each site.

Fifth, the implementation treats a physical track as exclusive in both directions: it checks a traversal against reservations for either direction. This is deliberately

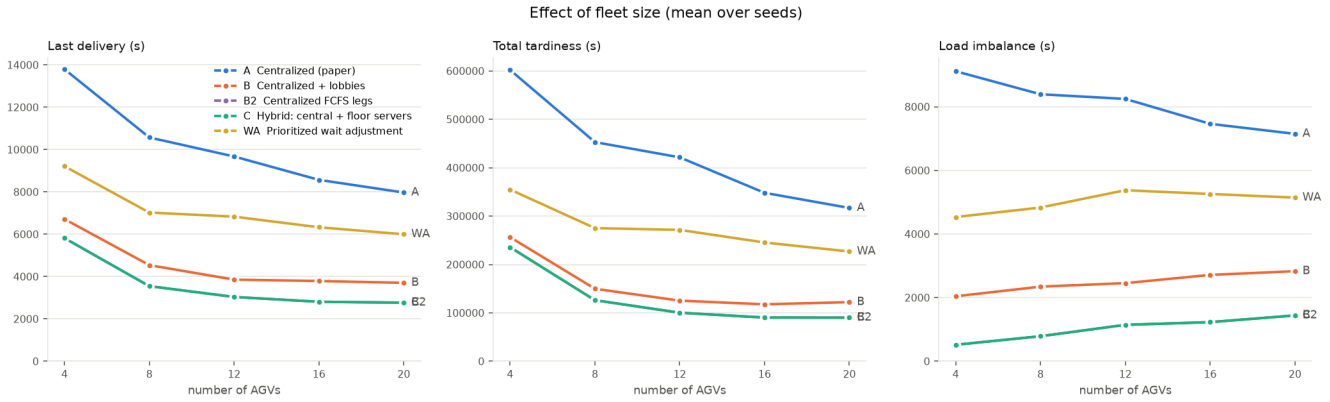


Fig. 5. Mean makespan, tardiness, and load imbalance by fleet size over the tested seeds.

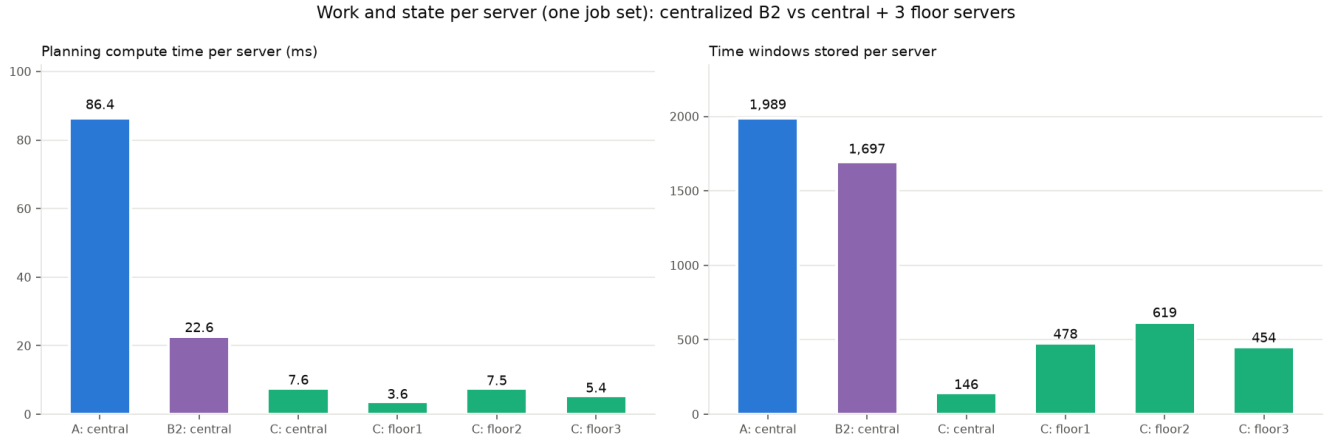


Fig. 6. Per-server computation time and reservation windows for centralized B2 and hybrid C on seed 7.

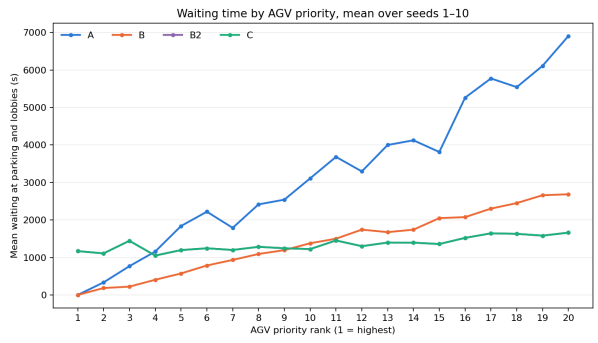


Fig. 7. Mean parking-plus-lobby wait by fixed AGV priority rank over seeds 1–10.

stricter than the paper's reverse-edge conflict check and can reject safe same-direction following schedules. It favors conservative separation but may overestimate congestion and completion time relative to facilities that permit controlled following.

Sixth, experiments use one hand-crafted factory graph and one synthetic job generator. This is a known generalizability limitation: topology, station placement, task mix, and due-date distributions may differ substantially in other plants. Finally, travel and lift times are deterministic functions of edge lengths and speed, with no uncertainty, acceleration, or congestion variation. The model also omits battery state and charging decisions. These are deliberate scope choices for isolating route conflicts, but actual travel-time variation and charging constraints can change feasible routes, resource contention, and the relative performance of the architectures.

IX. Conclusion

This paper presented a hybrid architecture for prioritized multi-AGV scheduling in a multi-floor smart-manufacturing layout. The work began with an implementation of the prioritized-planning baseline of Guo et al. (ICARCV 2024) [1], correcting the original demonstration's documented issues in conflict resolution, parking-node handling, service-time placement, and parameterization. The corrected baseline assigns jobs in earliest-pickup order using a fitness score, constructs shortest-path routes, and schedules AGVs by fixed priority while eliminating conflicts through sliding-window delays.

The main architectural contribution divides planning between a central server and one server per floor. The central server assigns jobs, separates routes into floor legs, and books shared lift trips. Floor servers construct conflict-free schedules for their own legs using local node and track reservations. Lift-lobby waiting points let an AGV wait near the floor-level conflict or lift that is delaying it. An independent checker validates each complete plan for route endpoints, node and bidirectional-track conflicts, edge travel times, dwell windows, and exactly-once job assignment. On the tested seeds, the hybrid planner in priority order reproduces the centralized planner with lobbies, providing a check that the route decomposition preserves that reference schedule under that processing order.

In the saved comparison of 100 jobs and 20 AGVs over ten seeds, the default FCFS hybrid configuration reduced mean last-delivery makespan by 65%, total tardiness by 72%, and load imbalance by 80% relative to the paper-style centralized configuration. The busiest-server CPU time fell by 92%, and the maximum number of reservation windows held by one server fell by 69%. These results demonstrate the potential value of lobby waiting and floor-level decomposition for the modeled three-floor factory. They are empirical results for the tested layout and settings, not a general guarantee of optimality or of the same improvement at larger scales [2].

The reduction in tardiness is substantial, but the mean total tardiness for the hybrid remains approximately 90,104.5 s across 100 jobs per seed. This indicates that conflict resolution and decomposition alone do not ensure that assigned deliveries meet their desired times. The current job assignment is greedy and priority-dependent, and delay accumulated by shared routes, lift use, and safety separation can still move deliveries well past their targets. Reducing this residual tardiness will require changes to assignment, sequencing, and resource coordination in addition to distributing the planning workload.

a) Future work.: Future work will extend the static, known-job setting to dynamic job arrivals and investigate battery limits and charging decisions. The lift model can be generalized to multiple lifts per floor, with lift capacity and service policies represented explicitly. A further architectural step is to remove the central lift-booking server and investigate fully decentralized lift coordination. Broader comparisons should include auction-based assignment, deep reinforcement learning (DRL), and conflict-based search (CBS), using common instances, objectives, and verification rules. The design should also be validated on physical multi-floor hardware.

Appendix A
Reproducibility and Implementation Map

The experiment outputs and implementation are available in the project repository [2]. The following map identifies the source locations for the algorithms described in the paper; line numbers refer to the repository version used to prepare this manuscript.

The regenerated tables use the CSV outputs under the experiment directories: results_hybrid_100j_20a_s1-10_strict/comparison.csv and sweep_agvs.csv; results_scale/sweep_jobs.csv; results_sensitivity/sensitivity.csv; results_baselines/assignment_baselines.csv; results_wait_adjustment/comparison.csv; results_lobby_capacity/capacity_sweep.csv; and the per-seed CSVs in results_fairness/. Every result folder is listed with the command, settings, and environment that produced it in results_manifest.json. Figures included with the paper are copies of the generated comparison and fairness plots.

From the repository root, install the project dependencies and run the experiments from the implementation directory:

```
cd agv_code/agv_code
pip install -r requirements.txt
python agv_compare.py --seeds 1-10 --jobs 100 --agvs 20
python sweep_job_scale.py
python sensitivity.py
python agv_baselines.py
python agv_compare.py --seeds 1-10 --jobs 100 --agvs 20 --sweep '' --outdir results_wait_adjustment
python lobby_capacity_sweep.py
python agv_fairness_experiment.py
```

Default output folders encode their settings (for example, results_hybrid_100j_20a_s1-10_strict/), so runs with different settings cannot overwrite each other; the CPU-repeat runs in Table 1 used –sweep '' with –outdir results_cpu_repeat_r1 to r5.

Compile from the paper/ directory with a LaTeX installation that includes IEEEtran and BibTeX:

```
pdflatex main.tex
bibtex main
pdflatex main.tex
pdflatex main.tex
```

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TABLE IX
Implementation map for the evaluated methods.

Component	Source location	Relevant implementation
Assignment and metrics	agv_code/agv_code/agv_core.py, lines 154–211 and 606–649	Fitness, route construction, Algorithm 1, reported metrics, and centralized pipeline.
Conflict scheduling	agv_code/agv_code/agv_core.py, lines 212–231, 268–408	Interval function, Algorithm 3, route planning, reservation freezing, and Algorithm 2.
Independent checking	agv_code/agv_code/agv_core.py, lines 409–567	Route continuity, service events, node/track/lift/lobby constraints, and verification.
Hybrid planning	agv_code/agv_code/agv_hybrid.py, lines 52–157 and 162–297	Route splitting, floor-leg planning, lift booking, finite-capacity admission, dispatch, and plan merging.
Layouts and comparison	agv_code/agv_code/agv_layouts.py, lines 47–90 and 106–137; agv_code/agv_code/agv_compare.py, lines 46–105 and 214–end	Three- and five-floor graphs, configuration runner, seeded comparison, and figures.
Additional experiments	agv_code/agv_code/sweep_job_scale.py; sensitivity.py; agv_baselines.py; agv_wait_adjust.py; agv_fairness_experiment.py; lobby_capacity_sweep.py	Job scale, parameter grid, assignment baselines, safe-interval planner, fairness variants, and lobby-capacity sweep.

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